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Inventor(s) MORI; Takaaki et al.

TRANSPORT MECHANISM AND ANALYSIS APPARATUS

Abstract

A transport mechanism includes: a transport table having a transport surface on which an analysis chip being transported; and a pair of screw shafts disposed parallel to each other and propelling the analysis chip in the transport direction, in which the transport surface has a hole formed in a contact region, and a recessed portion formed in at least a part of a periphery of an opening edge of the hole, and the recessed portion has a recessed surface that makes a height of a partial opening edge, which is located on a downstream side on an entire circumference of the opening edge and includes a contact point with a tangent extending in a direction orthogonal to the transport direction, lower than a height of the transport surface, and the recessed surface extends from the partial opening edge to an outside of the contact region.

Inventors: **MORI; Takaaki (Kanagawa, JP), DENAWA; Tatsuyuki (Kanagawa, JP)**

Applicant: **FUJIFILM CORPORATION** (Tokyo, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/JP2023/042080, filed on Nov. 22, 2023, which claims priority from Japanese Patent Application No. 2022-192254, filed on Nov. 30, 2022. The entire disclosure of each of the above applications is incorporated herein by reference.

BACKGROUND
1. Technical Field
[0002] The present disclosure relates to a transport mechanism and an analysis apparatus.
2. Related Art
[0003] A micro-total analysis system (TAS) is known as an analysis apparatus that analyzes a specimen such as a biological sample. The TAS is an analysis apparatus that uses an analysis chip comprising a micro flow channel to perform a series of chemical or biochemical analysis steps within the analysis chip, such as mixing of a specimen and a reagent, a chemical reaction or a biochemical reaction, and detection of a product after the reaction. The analysis chip is loaded into the analysis apparatus, and various processes are executed within the analysis apparatus, such as a specimen dispensing process of dispensing a specimen, a reagent dispensing process of dispensing a reagent, a process for causing a specimen and a reagent to react with each other, and an optical detection process. The analysis chip is transported to each processing unit that executes various processes by a transport mechanism provided in the analysis apparatus.
[0004] As a transport mechanism that transports an object to be transported, a configuration is generally known in which the object to be transported is placed on a transport belt, the transport belt is rotated, and the object to be transported is transported for each transport belt, as in a belt conveyor. With the belt conveyor, the object to be transported is stationarily placed on the transport belt and does not move relative to the transport belt, so that the object to be transported can be stably transported. On the other hand, since the belt conveyor needs to be rotated, a structure such as a space for rotating the transport belt and a mechanism for driving the transport belt is required on a back surface of a transport passage, and space saving is difficult.
[0005] As a space-saving transport device, a transport device that comprises a screw shaft that includes a shaft body disposed along a transport direction and a ridge helically disposed on an outer periphery of the shaft body, and that transports an object to be transported by rotating the screw shaft is disclosed in JP2013-184801A.
[0006] For example, the transport device of JP2013-184801A holds a holder between a pair of screw shafts disposed in parallel, and sends the holder along a transport path to transport a specimen container held by the holder.
SUMMARY
[0007] The transport device of JP2013-184801A transports a holder that holds a specimen container accommodating a specimen in an upright state, and the specimen container is assumed to be a blood collection tube.
[0008] On the other hand, the analysis chip for TAS is, for example, a microchannel device comprising a microchannel composed of a resin plate having a groove formed on one surface and a resin film to cover the groove, and has a shape significantly different from that of a test container such as the blood collection tube assumed in JP2013-184801A. Therefore, it is difficult to directly apply the transport device of JP2013-184801A to transporting an analysis chip having a flat bottom

surface, such as the analysis chip for TAS.
[0009] The present inventors are considering a transport mechanism that comprises a pair of screw shafts and that is configured to transport an analysis chip having a flat bottom surface by sliding it on a transport table. In such a transport mechanism, in a case of transporting an analysis chip having a flat bottom surface along the transport table, it has become apparent that in a case in which there are holes for screw insertion to install the transport table on a body of the apparatus present on a surface of the transport table, the analysis chip may get caught in the holes and become damaged.
[0010] An object of the present disclosure is to provide a transport mechanism and an analysis apparatus capable of stably transporting an analysis chip having a flat bottom surface and suppressing damage to the analysis chip.
[0011] According to the present disclosure, there is provided a transport mechanism that transports an analysis chip including a body part having a flat bottom surface and a pair of engagement ribs provided to protrude outward from both ends in a width direction of the body part in a transport direction orthogonal to the width direction, the transport mechanism comprising: a transport table having a transport surface on which the analysis chip is placed with the bottom surface in contact with the transport surface, the analysis chip being transported while sliding on the transport surface; and a pair of screw shafts disposed parallel to each other along the transport direction, each of which has a shaft body and a helical protrusion helically formed on an outer peripheral surface of the shaft body, and the pair of screw shafts propels the analysis chip in the transport direction by rotating about their respective axes while the helical protrusions are engaged with the engagement ribs, in which the transport surface has a hole formed in a contact region where the bottom surface of the analysis chip contacts the transport surface, and a recessed portion formed in at least a part of a periphery of an opening edge of the hole, and the recessed portion has a recessed surface that makes a height of a partial opening edge, which is located on a downstream side in the transport direction on an entire circumference of the opening edge and includes a contact point with a tangent extending in a direction orthogonal to the transport direction, lower than a height of the transport surface, and the recessed surface extends from the partial opening edge to an outside of the contact region.
[0012] It is preferable that a ratio of a diameter of the hole to a length in the width direction of the body part is 8% or more, and the diameter is less than a length in the width direction of the body part.
[0013] It is preferable that a diameter of the hole is 4 mm or more.
[0014] It is preferable that the analysis chip has a planar shape in which a front end portion located on the downstream side in the transport direction has a front edge formed by a straight line extending in the direction orthogonal to the transport direction.
[0015] It is preferable that a partial outer edge located on the downstream side in the transport direction on an entire circumference of an outer edge of the recessed portion and extending to the outside of the contact region extends toward the outside of the contact region in an inclined manner toward the downstream side in the transport direction with respect to the tangent, and the partial outer edge is a straight line or has an arc shape convex toward the outside of the contact region.
[0016] It is preferable that an overall planar shape of the recessed portion including the hole is an oval shape of which a minor diameter is a diameter of the hole and one side in a longitudinal direction is the partial outer edge, and an entirety of a semicircular portion on an opposite side of the hole in the oval shape is located outside the contact region.
[0017] It is preferable that a plurality of the holes are provided, and the recessed portion is provided for each of the holes.
[0018] It is preferable that the analysis chip is a microchannel device including a microchannel for analyzing a specimen consisting of a biological sample.
[0019] According to the present disclosure, there is provided an analysis apparatus comprising: the

transport mechanism according to the present disclosure; and a plurality of processing units disposed along the transport direction of the transport table and performing predetermined processing on a specimen accommodated in the analysis chip.
[0020] With the transport mechanism and the analysis apparatus of the technology of the present disclosure, it is possible to stably transport the analysis chip having a flat bottom surface and suppress damage to the analysis chip.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0021] FIG. **1** is a schematic view showing a schematic configuration of an analysis apparatus.
[0022] FIG. **2** is a perspective view of an analysis chip with a bottom surface facing up.
[0023] FIG. **3** is an exploded perspective view of the analysis chip.
[0024] FIG. **4** is a perspective view of the analysis chip shown in FIG. **2** with its top and bottom reversed.
[0025] FIG. **5** is a perspective view showing a schematic configuration of a transport mechanism.
[0026] FIG. **6** is an enlarged plan view showing a part of a transport table on which the analysis chip is placed.
[0027] FIG. **7** is an enlarged perspective view showing a part of a transport table on which the analysis chip is placed.
[0028] FIG. **8** is an explanatory diagram of a modification example of a recessed portion.
[0029] FIG. **9** is an enlarged plan view showing a part of the transport mechanism.
[0030] FIG. **10** is an explanatory diagram of a problem that occurs in a case in which the transport table comprises a hole that does not comprise a recessed portion.
DESCRIPTION OF EMBODIMENTS
[0031] Hereinafter, a transport mechanism and an analysis apparatus comprising the transport mechanism according to one embodiment of the present disclosure will be described with reference to the drawings.
[0032] FIG. **1** is a schematic view showing a schematic configuration of an analysis apparatus comprising a transport mechanism. An analysis apparatus **10** is, for example, an immunological analysis apparatus that detects a test target substance in a specimen collected from a living body by utilizing an antigen-antibody reaction. The analysis apparatus **10** dispenses a specimen and a reagent to an analysis chip **12**, causes mixing and an immune reaction in a microchannel in the analysis chip **12**, optically detects the test target substance in the specimen, and outputs a test result.
[0033] The analysis chip **12** is a microchannel device comprising a microchannel **32** therein (see FIG. **2**), and is, for example, an analysis chip for TAS. As an example, a reagent such as a labeled antibody solution and a buffer solution is introduced into the microchannel **32** in addition to the specimen. In the microchannel **32**, an immune complex containing the test target substance in the specimen is generated and bound/free (B/F) separated by isotachophoresis, and separation and fractionation are performed by capillary electrophoresis. Then, the immune complex is irradiated with excitation light, and the fluorescence intensity corresponding to the concentration of the test target substance in the specimen is measured. The specimen is, for example, a biological sample such as blood collected from a living body. First, details of the analysis chip **12** will be described.
[Analysis Chip]
[0034] FIG. **2** is a perspective view of the analysis chip **12** with a bottom surface **12**Aa facing up. FIG. **3** is an exploded perspective view of the analysis chip **12**. FIG. **4** is a perspective view of the analysis chip **12** shown in FIG. **2** with its top and bottom reversed.
[0035] As shown in FIG. **2**, the analysis chip **12** comprises a body part **12**A and a pair of

engagement ribs **12**B. In the body part **12**A, one surface **12**Aa is flat. For convenience of description, in FIG. **2**, the analysis chip **12** is shown with the bottom surface **12**Aa facing up. In a case in which the analysis chip **12** is transported by the transport mechanism **14**, the analysis chip **12** is transported with the one surface **12**Aa serving as the bottom surface (see FIG. **4**). Accordingly, this one surface **12**Aa is referred to as the bottom surface **12**Aa of the body part **12**A of the analysis chip **12**. That is, the analysis chip **12** comprises the body part **12**A of which the bottom surface **12**Aa is flat. The pair of engagement ribs **12**B are provided to protrude outward from both ends in a width direction of the body part **12**A. Here, the width direction of the body part **12**A is a direction orthogonal to a transport direction A of the analysis chip **12**. In addition, the analysis chip **12** comprises a pair of support ribs **12**C that serve as a support portion in a case in which the analysis chip **12** is placed in a posture shown in FIG. **2**. The support ribs **12**C are provided to protrude outward from both ends in the width direction of the body part **12**A, as with the engagement rib **12**B, but a protruding width is smaller than that of the engagement rib **12**B. The analysis chip **12** comprises the microchannel **32** and a plurality of wells **33** for dispensing a specimen, a reagent, and the like in the microchannel **32**.
[0036] As shown in FIG. **3**, the analysis chip **12** is composed of a resin plate **30** and a film **31**.
[0037] The resin plate **30** is provided with a groove **34** constituting the microchannel **32** (see FIG. **2**) and the plurality of wells **33**. The microchannel **32** is, for example, a flow channel having a width of several tens of μm, and the groove **34** is a groove having a width of several tens of μm. The well **33** consists of a through-hole **33***a* that penetrates from one surface **30***a* of the resin plate **30** to the other surface **30***b* which is a back surface thereof, and a cylindrical portion **33***b* where an opening peripheral edge of the through-hole **33***a* protrudes from a surface on the other surface **30***b* side.
[0038] The film **31** is bonded to the one surface **30***a* of the resin plate **30** on which the groove **34** and the well **33** are formed so as to cover the groove **34** and the through-hole **33***a* (see FIGS. **2** and **3**). As a result, a portion of the groove **34** and the through-hole **33***a* that is open on the one surface **30***a* is sealed. The one surface **30***a* of the resin plate **30** to which the film **31** is bonded corresponds to the bottom surface **12**Aa of the analysis chip **12** (see FIG. **2**).
[0039] The pair of engagement ribs **12**B and support ribs **12**C described above are formed on the resin plate **30**. In addition, the resin plate **30** comprises a through-hole **36** that is separate from the through-hole **33***a* of the well **33** and that penetrates the resin plate **30** in an up-down direction in an end part region of the body part **12**A adjacent to the engagement rib **12**B. The through-hole **36** is a hole through which a positioning pin **62** of the transport mechanism **14**, which will be described below, is inserted. In this example, the through-holes **36** are provided at two positions. The through-hole **36** is provided at a position adjacent to the engagement rib **12**B in the end part region in the width direction of the body part **12**A.
[0040] As shown in FIG. **4**, the analysis chip **12** is transported in the transport direction A orthogonal to the width direction with the film **31** side as a bottom surface. In addition, the analysis chip **12** is transported in a position in which the engagement rib **12**B is at a front end located on a downstream side in the transport direction A, and the support rib **12**C is at a rear end located on an upstream side in the transport direction A. The analysis chip **12** has a planar shape in which a front end portion **12**E located on the downstream side in the transport direction A has a front edge a formed by a straight line extending along the direction orthogonal to the transport direction A (see FIG. **6**).
[0041] Returning to FIG. **1**, a configuration of the analysis apparatus **10** will be described. The analysis apparatus **10** comprises, for example, a stocker **13** that accommodates a plurality of the analysis chips **12**, the transport mechanism **14** that transports the analysis chip **12**, a plurality of processing units **15** to **19** that perform predetermined processing on a specimen accommodated in the analysis chip **12**, a chip disposal unit **20**, and a processor **21** that functions as a control unit.
[0042] The transport mechanism **14** comprises a transport table **40** extending along the transport

direction A of the analysis chip **12** and a pair of screw shafts **50** and **52** (see FIG. **5**). Although details of the transport mechanism **14** will be described below, the transport table **40** has a plurality of stop positions P**1** to P**5** where the analysis chip **12** is stopped. In addition, in this example, the transport mechanism **14** comprises a positioning mechanism **60** that positions the analysis chip **12**, which is stopped at the stop positions P**1** to P**5**, at preset positions in the stop positions P**1** to P**5**. The positioning mechanism **60** comprises a positioning pin **62** that positions the analysis chip **12** by inserting its tip into a positioning hole provided at each of the stop positions P**1** to P**5** of the transport table **40**, and a lifting mechanism **64** that lifts and lowers the positioning pin **62**. The lifting mechanism **64** lifts the positioning pin **62** between an insertion position where the positioning pin **62** is inserted through the through-hole **36** of the analysis chip **12** and the tip is inserted into the positioning hole and a retreat position where the positioning pin **62** is moved upward from the insertion position and is retreated from the insertion position. The lifting mechanism **64** is, for example, an eccentric cam.

[0043] The plurality of analysis chips **12** are stored in the stocker **13** in advance. The analysis chip **12** is taken out from the stocker **13** by a chipset mechanism (not shown) and is set on the transport table **40**. The stocker **13** is disposed below an upstream end of the transport table **40**, and the analysis chip **12** passes through a chipset opening **48** provided at the upstream end of the transport table **40** and is set on a transport surface **42** (see FIG. **5**).

[0044] The plurality of processing units **15** to **19** are, for example, a specimen dispensing processing unit **15**, a reagent dispensing processing unit **16**, a pressurization processing unit **17**, a reaction processing unit **18**, and a detection processing unit **19**. The plurality of processing units **15** to **19** are disposed to correspond to the respective stop positions P**1** to P**5** along the transport direction A of the analysis chip **12**.

[0045] The specimen dispensing processing unit **15** comprises a dispensing mechanism **15***b* comprising a nozzle **15***a*, and dispenses a specimen into the microchannel **32** of the analysis chip **12** stopped at the stop position P**1**. A specimen collection tube (not shown) accommodating the collected specimen is loaded into the analysis apparatus **10**, and the nozzle **15***a* sucks the specimen from the specimen collection tube and dispenses the specimen into the microchannel **32** from the predetermined well **33** of the analysis chip **12**.

[0046] The reagent dispensing processing unit **16** comprises a dispensing mechanism **16***b* comprising a nozzle **16***a*, and dispenses a reagent into the microchannel **32** of the analysis chip **12** stopped at the stop position P**2**. A reagent container (not shown) accommodating the reagent is loaded into the analysis apparatus **10**, and the nozzle **16***a* sucks the reagent from the reagent container and dispenses the reagent into the microchannel **32** from the predetermined well **33** of the analysis chip **12**.

[0047] In the pressurization processing unit **17**, gas is fed through at least a part of the plurality of wells **33** from the outside of the analysis chip **12** to introduce the specimen and the reagent into the microchannel **32** and mix them, so that the inside of the microchannel **32** is pressurized.

[0048] In the reaction processing unit **18**, a labeled antibody is concentrated, an antigen-antibody reaction is performed, and B/F separation is performed by isotachophoresis.

[0049] In the detection processing unit **19**, the separation and fractionation of the immune complex are performed by capillary gel electrophoresis method, and then detection processing of detecting the test target substance in the specimen is executed. For example, the test target substance contained in the immune complex is detected by a laser fluorescence method. The detection processing unit **19** comprises an optical head **19***a*. The optical head **19***a* is disposed at a position where fluorescence from a predetermined location of the microchannel **32** can be received while irradiating the predetermined location with laser light from the bottom surface **12**Aa of the analysis chip **12** in a state where the analysis chip **12** is stopped at the stop position P**5**. The optical head **19***a* comprises, for example, an irradiation section and a light receiving section.

[0050] The analysis apparatus **10** further comprises a pneumatic section (not shown) that pressurizes and depressurizes the inside of the flow channel and an electrophoresis section **22** for electrophoresis. During the pressurization processing, the reaction processing, and the detection processing, a pressing unit **24** comprising a sealing portion and a supply pipe presses a surface of the analysis chip **12**, which comprises the well **33**, in order to seal the well **33** or to connect a gas supply pipe to the well **33**. The pressing unit **24** can be lifted and lowered by the lifting mechanism between a position where it is separated from the analysis chip **12** and a position where it presses the analysis chip **12**. In this example, the lifting mechanism **64** provided in the positioning mechanism **60** of the transport mechanism **14** also serves as the lifting mechanism of the pressing unit **24**.

[0051] The chip disposal unit **20** accommodates the analysis chip **12** after the analysis has been completed. The chip disposal unit **20** is disposed at a location where the analysis chip **12** that falls from a disposal opening **49** provided at a downstream end of the transport table **40** is received.

[0052] The processor **21** constitutes a control unit that integrally controls each unit of the analysis apparatus **10**. An example of the processor **21** is a central processing unit (CPU) which performs various types of control by executing a program. The CPU functions as a control unit which controls each unit by executing a program. The processor **21** also serves as a control unit of the transport mechanism **14**. The processor **21** functions as a control unit of the transport mechanism **14**, and controls a driving unit **54** of the transport mechanism **14**, which will be described below. As a result, the processor controls rotation of the screw shafts **50** and **52** to control movement of the analysis chip **12** between the plurality of stop positions P**1** to P**5** and stopping of the analysis chip **12** at the plurality of stop positions P**1** to P**5**.

[0053] Hereinafter, the transport mechanism **14** will be described in detail.

[Transport Mechanism]

[0054] FIG. **5** is a perspective view showing a schematic configuration of the transport mechanism **14**. The transport mechanism **14** transports the above-described analysis chip **12** in the transport direction A orthogonal to the width direction of the analysis chip **12**. The transport mechanism **14** comprises the transport table **40** and the pair of screw shafts **50** and **52**.

[0055] FIG. **6** is an enlarged plan view showing a part of the transport table **40** on which the analysis chip **12** is placed, and FIG. **7** is an enlarged perspective view showing a part of the transport table **40** on which the analysis chip **12** is placed. In FIGS. **6** and **7**, the screw shafts **50** and **52** disposed on the transport table **40** are omitted.

[0056] The transport table **40** has the transport surface **42** on which the bottom surface **12**Aa of the analysis chip **12** is placed with the bottom surface **12**Aa in contact with the transport surface **42**. The analysis chip **12** is transported by sliding on the transport surface **42** of the transport table **40** in a state where the bottom surface **12**Aa of the analysis chip **12** is in contact with the transport surface **42**. A temperature of the transport surface **42** may be controlled by temperature control means (not shown). The temperature control means comprises, for example, a heater and a temperature control circuit. In a case in which the analysis chip **12** is transported with the bottom surface **12**Aa in contact with the temperature-controlled transport surface **42**, the specimen and the reagent accommodated in the analysis chip **12** can be transported while being temperature-controlled. A region of the transport surface **42** with which the bottom surface **12**Aa of the analysis chip **12** comes into contact in a case of transporting the analysis chip **12** is referred to as a contact region **42**A. The contact region **42**A is a region between the pair of screw shafts **50** and **52**, and is a region that extends along the transport direction A with a width equivalent to a length L in the width direction of the body part **12**A of the analysis chip **12**. In FIGS. **5** to **7** and the like, a region between two-dot chain lines **41** on the transport surface **42** corresponds to the contact region **42**A.

[0057] The transport surface **42** has a hole **43** formed in the contact region **42**A and a recessed portion **45** formed in at least a part of a periphery of an opening edge **44** of the hole **43**. The hole **43** formed in the contact region **42**A of the transport surface **42** is a positioning hole into which the tip of the positioning pin **62** is inserted, a hole for fixing screw insertion for fixing the transport table

40 in the analysis apparatus **10**, or the like.
[0058] The recessed portion **45** has a recessed surface **46** that makes a height of a partial opening edge **44**_a_, which is located on the downstream side in the transport direction A on the entire circumference of the opening edge **44** and includes a contact point C with a tangent B extending in the direction orthogonal to the transport direction A, lower than the transport surface **42**, and the recessed surface **46** extends from the partial opening edge **44**_a_ to an outside of the contact region **42**A.
[0059] The hole **43** comprising the recessed portion **45** is a round hole as an example. A ratio of a diameter φ of the hole **43** to a length L in the width direction of the body part **12**A of the analysis chip **12** is, as an example, 8% or more, and the diameter φ is less than the length L. In a more specific example, the diameter φ of the hole **43** is 4 mm or more.
[0060] A partial outer edge **47**_a_ located on the downstream side in the transport direction A on an entire circumference of an outer edge **47** of the recessed portion **45** and extending to the outside of the contact region **42**A extends toward the outside of the contact region **42**A in an inclined manner toward the downstream side in the transport direction A with respect to the tangent B. The partial outer edge **47**_a_ is a straight line in this example. In addition, as shown in FIG. **8** as a modification example, the partial outer edge **47**_a_ may have an arc shape convex toward the outside of the contact region **42**A.
[0061] In this example, an overall planar shape of the recessed portion **45** including the hole **43** is an oval shape of which a minor diameter is a diameter of the hole **43** and one side in a longitudinal direction is the partial outer edge **47**_a_. In the oval shape, the entirety of a semicircular portion **47**_b_ on an opposite side of the hole **43** is located outside the contact region **42**A.
[0062] As shown in FIG. **5**, in this example, a plurality of the holes **43** are formed on the transport surface **42**, and the recessed portion **45** is provided for each of the holes **43**.
[0063] As shown in FIG. **5**, the pair of screw shafts **50** and **52** are disposed on the transport table **40**. The pair of screw shafts **50** and **52** are disposed parallel to each other along the transport direction A. The screw shaft **50** has a shaft body **50**A and a helical protrusion **50**B helically formed on an outer peripheral surface of the shaft body **50**A. Similarly, the screw shaft **52** has a shaft body **52**A and a helical protrusion **52**B helically formed on an outer peripheral surface of the shaft body **52**A.
[0064] A helical pitch SP of the helical protrusion **50**B of one screw shaft **50** and a helical pitch SP of the helical protrusion **52**B of the other screw shaft **52** are the same (see FIG. **9**). A direction of helical rotation of the helical protrusion **50**B of one screw shaft **50** is opposite to a direction of helical rotation of the helical protrusion **52**B of the other screw shaft **52**. One end of each of the pair of screw shafts **50** and **52** is connected to the driving unit **54** that rotationally drives the screw shafts **50** and **52**. The pair of screw shafts **50** and **52** are attached to the driving unit **54** such that phases of the helical protrusions **50**B and **52**B match each other. The driving unit **54** includes, for example, a motor **55** and a plurality of gears (not shown). One end of one screw shaft **50** is connected to the motor **55** such that a driving force of the motor **55** can be directly transmitted as a rotational force. One end of the other screw shaft **52** is connected to one of the plurality of gears, and the driving force of the motor **55** is transmitted as a rotational force via the plurality of gears. The plurality of gears are combined such that a rotation direction of one screw shaft **50** and a rotation direction of the other screw shaft **52** are opposite to each other. The other end of each of the screw shafts **50** and **52** is supported by a bearing unit **56**.
[0065] The pair of screw shafts **50** and **52** rotate about their axes while the respective helical protrusions **50**B and **52**B and the pair of engagement ribs **12**B are engaged with each other, thereby propelling the analysis chip **12** in the transport direction A.
[0066] FIG. **9** is an enlarged plan view showing a part of the transport mechanism **14**. The analysis chip **12** is placed on the transport surface **42** of the transport table **40** in a posture in which the flat bottom surface **12**Aa (see FIG. **2**) is in contact with the transport surface **42** and the front end

portion **12**E extending in the width direction is orthogonal to the transport direction A. In addition, the analysis chip **12** is placed between the pair of screw shafts **50** and **52** disposed parallel to each other. The helical protrusion **50**B is a series of convex portions helically formed along an axial direction, and a protrusion portion **50**Ba is defined as one convex portion in a case in which the helical protrusion **50**B is viewed in a plan view. A plurality of the protrusion portions **50**Ba are arranged at equal pitches SP along the axial direction in a case in which the helical protrusion **50**B is viewed in a plan view. Similarly, a protrusion portion **52**Ba is defined as one convex portion in a case in which the helical protrusion **52**B is viewed in a plan view. A plurality of the protrusion portions **52**Ba are arranged at equal pitches SP along the axial direction in a case in which the helical protrusion **52**B is viewed in a plan view. In the analysis chip **12**, one engagement rib **12**B is located in a recess portion between two protrusion portions **50**Ba adjacent to each other in the axial direction in a case in which a helical groove of the screw shaft **50**, more specifically, the helical protrusion **50**B is viewed in a plan view. Further, the other engagement rib **12**B is placed to be located in a recess portion between two protrusion portions **52**Ba adjacent to each other in the axial direction in a case in which a helical groove of the screw shaft **52**, more specifically, the helical protrusion **52**B is viewed in a plan view. An interval of the recess portion between the two adjacent protrusion portions **50**Ba is narrower than the pitch SP of the helical protrusion **50**B by a width of one protrusion portion **50**Ba, but is substantially equal to the pitch SP. An interval between the two adjacent protrusion portions **52**Ba is also narrower than the pitch SP of the helical protrusion **52**B by a width of one protrusion portion **52**Ba, but is substantially equal to the pitch SP.

[0067] The engagement between the screw shafts **50** and **52** and the pair of engagement ribs **12**B will be described in more detail below. The engagement between the screw shaft **50** and one engagement rib **12**B will be described as an example. In a case in which the helical protrusion **50**B rotates due to the rotation of the screw shaft **50**, a phase of the protrusion portion **50**Ba, which is a convex portion of the helical protrusion **50**B, and a phase of the recess portion, which is a groove of the adjacent protrusion portion **50**Ba, move in the transport direction. As a result, the protrusion portion **50**Ba presses the engagement rib **12**B from the rear (upstream side), and propels the engagement rib **12**B in the axial direction of the screw shaft **50** in accordance with the change in phase of the protrusion portion **50**Ba while maintaining a contact state with the engagement rib **12**B. The same applies to the engagement between the screw shaft **52** and the other engagement rib **12**B. In this way, the screw shafts **50** and **52** are rotated about their axes while the helical protrusions **50**B and **52**B and the engagement ribs **12**B and **12**B are engaged with each other, whereby the engagement ribs **12**B and **12**B are pressed in the axial direction by the helical protrusions **50**B and **52**B. As a result, the analysis chip **12** is transported in the transport direction A which is the axial direction. In this case, the analysis chip **12** is transported while sliding on the transport surface **42** in a state where the bottom surface **12**Aa is in contact with the transport surface **42**.

[0068] As described above, the present transport mechanism **14** comprises the transport table **40** having the transport surface **42** on which the analysis chip **12** is transported while sliding and the pair of screw shafts **50** and **52** that propel the analysis chip **12** in the transport direction A by rotating about their axes while the helical protrusions **50**B and **52**B and the engagement ribs **12**B and **12**B are engaged with each other. With the present transport mechanism **14**, by rotating the pair of screw shafts **50** and **52**, the analysis chip **12** is transported in such a manner that it slides on the transport surface **42** of the transport table **40**, thereby enabling stable transport of the analysis chip **12**. In addition, since the screw shafts **50** and **52** are rotated to transport the analysis chip, space saving can be achieved compared to a transport mechanism that requires a driving unit on a back surface side of a transport table as in a belt conveyor. In addition, compared to a belt conveyor, it is possible to improve the positional accuracy in a case of stopping the analysis chip **12** at a desired stop position.

[0069] As described above, the hole **43** such as a positioning hole and a hole for fixing screw insertion is formed on the transport surface **42**. In the present embodiment, the recessed portion **45** is formed in at least a part of the periphery of the opening edge **44** of the hole **43**, but, in the absence of such a recessed portion **45**, the following problem arises. Referring to FIG. **10**, the problem will be described. In FIG. **10**, the same components as those in the transport mechanism **14** of the present embodiment are denoted by the same reference numerals.

[0070] In FIG. **10**, the transport table **40** does not have the recessed portion **45** in the hole **43** of the transport surface **42**. In this case, in a case in which the flat bottom surface **12**Aa is slid over the contact region **42**A, the front end portion **12**E located on the downstream side of the analysis chip **12** may get caught on the partial opening edge **44***a*, which is located on the downstream side in the transport direction A of the opening edge **44** of the hole **43**. In particular, the front end portion **12**E of the analysis chip **12** is likely to get caught on the contact point C with a tangent extending in the direction orthogonal to the transport direction A of the partial opening edge **44***a*. In particular, the analysis chip **12** in this example is configured such that, as shown in FIG. **5**, the bottom surface **12**Aa is formed by the film **31** bonded to the resin plate **30**, so that an end part of the film **31** is likely to get caught on the partial opening edge **44***a* of the hole **43**. In a case in which the film **31** gets caught on the partial opening edge **44***a* of the hole **43**, the film **31** may peel off from the resin plate **30**, leading to damage to the analysis chip **12**.

[0071] On the other hand, in the present transport mechanism **14**, the transport surface **42** has the recessed portion **45** formed in at least a part of the periphery of the opening edge **44** of the hole **43**, and this recessed portion **45** has the recessed surface **46** that makes the height of the partial opening edge **44***a*, which includes the contact point C with the tangent B extending in the direction orthogonal to the transport direction A on the entire circumference of the opening edge **44**, lower than the height of the transport surface **42**. The recessed surface **46** extends from the partial opening edge **44***a* to the outside of the contact region **42**A. Therefore, in the contact region **42**A, there is no portion orthogonal to the transport direction A and flush with the transport surface **42** at the opening edge **44** of the hole **43** or the downstream edge of the recessed portion **45** provided on the transport surface **42**. Thus, the analysis chip **12** can be prevented from getting caught in the hole **43** or the recessed portion **45** during the transport, and as a result, damage to the analysis chip **12** can be suppressed.

[0072] In a case in which the ratio of the diameter φ of the hole **43** to the length L in the width direction of the analysis chip **12** is 8% or more, the analysis chip **12** is likely to get caught during the transport. Therefore, in a case in which the ratio of the diameter of the hole **43** to the length L in the width direction of the analysis chip **12** is 8% or more, the effect of providing the recessed portion **45** described above is significant.

[0073] In the case of using μTAS, the length L in the width direction of the analysis chip **12** is often 40 mm to 50 mm, and, in a case of using the analysis chip **12** of this size, in a case in which the diameter φ of the hole **43** is 4 mm or more, the analysis chip **12** is likely to get caught during the transport. Therefore, in a case in which the diameter of the hole **43** is 4 mm or more, the effect of providing the recessed portion **45** described above is significant.

[0074] In a case in which there are a plurality of holes **43** in the contact region **42**A of the transport surface **42**, it is preferable that each hole **43** is provided with the recessed portion **45**. By providing the recessed portion **45** in each hole **43**, it is possible to prevent the analysis chip **12** from getting caught in each hole **43**.

[0075] On the transport surface **42**, there may be holes that do not have the recessed portion **45**. In particular, in a case in which the ratio of the diameter of the hole to the length L in the width direction of the body part **12**A of the analysis chip **12** is less than 8%, the analysis chip **12** is less likely to get caught, so that such a hole with that diameter does not need to be provided with the recessed portion **45**. For example, in a case in which the transport table **40** is a transport table for transporting the analysis chip **12** with L being about 40 mm to 50 mm, a hole having a diameter less than 4 mm does not need to be provided with the recessed portion **45**.

[0076] The analysis apparatus **10** comprises the transport mechanism **14**, which can suppress damage to the analysis chip **12**. Therefore, the occurrence of an error due to the damage to the analysis chip **12** during the analysis can be suppressed, and the analysis efficiency can be improved.
[0077] The transport mechanism **14** can be applied to an analysis apparatus other than the analysis apparatus **10** described above, and the analysis chip that can be transported by the transport mechanism **14** is not limited to the analysis chip **12**. The analysis chip transported by the transport mechanism **14** may be any analysis chip as long as it comprises at least a body part having a flat bottom surface and a pair of engagement ribs provided to protrude outward from both ends in a width direction of the body part.
[0078] In addition, in the above-described embodiment, as a hardware structure of the processor, various processors shown below can be used. The various processors include, in addition to a CPU that is a general-purpose processor that executes software (program) to function as various processing units, a programmable logic device (PLD) of which a circuit configuration can be changed after manufacturing, such as a field-programmable gate array (FPGA), and a dedicated electric circuit that is a processor having a circuit configuration dedicatedly designed for executing specific processing, such as an application specific integrated circuit (ASIC).
[0079] The above-described processing may be executed by one of these various processors, or a combination of two or more processors of the same type or different types (for example, a combination of a plurality of FPGAs and a combination of a CPU and an FPGA). In addition, a plurality of processing units may be configured of one processor. As an example in which the plurality of processing units are configured of one processor, there is a form in which a processor that realizes all functions of a system including the plurality of processing units by using one integrated circuit (IC) chip is used, such as a system on chip (SOC).
[0080] Furthermore, the hardware structure of the processors is, more specifically, an electric circuit (circuitry) in which circuit elements such as semiconductor elements are combined.
[0081] The following appendices are further disclosed with respect to the above embodiment.
(Appendix 1)
[0082] A transport mechanism that transports an analysis chip including a body part having a flat bottom surface and a pair of engagement ribs provided to protrude outward from both ends in a width direction of the body part in a transport direction orthogonal to the width direction, the transport mechanism comprising: [0083] a transport table having a transport surface on which the analysis chip is placed with the bottom surface in contact with the transport surface, the analysis chip being transported while sliding on the transport surface; and [0084] a pair of screw shafts disposed parallel to each other along the transport direction, each of which has a shaft body and a helical protrusion helically formed on an outer peripheral surface of the shaft body, and the pair of screw shafts propels the analysis chip in the transport direction by rotating about their respective axes while the helical protrusions are engaged with the engagement ribs, [0085] in which the transport surface has a hole formed in a contact region where the bottom surface of the analysis chip contacts the transport surface, and a recessed portion formed in at least a part of a periphery of an opening edge of the hole, and [0086] the recessed portion has a recessed surface that makes a height of a partial opening edge, which is located on a downstream side in the transport direction on an entire circumference of the opening edge and includes a contact point with a tangent extending in a direction orthogonal to the transport direction, lower than a height of the transport surface, and the recessed surface extends from the partial opening edge to an outside of the contact region.
(Appendix 2)
[0087] The transport mechanism according to Appendix 1, [0088] in which a ratio of a diameter of the hole to a length in the width direction of the body part is 8% or more, and the diameter is less than a length in the width direction of the body part.

(Appendix 3)
[0089] The transport mechanism according to Appendix 1 or 2, [0090] in which a diameter of the hole is 4 mm or more.
(Appendix 4)
[0091] The transport mechanism according to any one of Appendices 1 to 3, [0092] in which the analysis chip has a planar shape in which a front end portion located on the downstream side in the transport direction has a front edge formed by a straight line extending in the direction orthogonal to the transport direction.
(Appendix 5)
[0093] The transport mechanism according to any one of Appendices 1 to 4, [0094] in which a partial outer edge located on the downstream side in the transport direction on an entire circumference of an outer edge of the recessed portion and extending to the outside of the contact region extends toward the outside of the contact region in an inclined manner toward the downstream side in the transport direction with respect to the tangent, and [0095] the partial outer edge is a straight line or has an arc shape convex toward the outside of the contact region.
(Appendix 6)
[0096] The transport mechanism according to Appendix 5, [0097] in which an overall planar shape of the recessed portion including the hole is an oval shape of which a minor diameter is a diameter of the hole and one side in a longitudinal direction is the partial outer edge, and [0098] an entirety of a semicircular portion on an opposite side of the hole in the oval shape is located outside the contact region.
(Appendix 7)
[0099] The transport mechanism according to any one of Appendices 1 to 6, [0100] in which a plurality of the holes are provided, and the recessed portion is provided for each of the holes.
(Appendix 8)
[0101] The transport mechanism according to any one of Appendices 1 to 7, [0102] in which the analysis chip is a microchannel device including a microchannel for analyzing a specimen consisting of a biological sample.
(Appendix 9)
[0103] An analysis apparatus comprising: [0104] the transport mechanism according to any one of Appendices 1 to 8; and [0105] a plurality of processing units disposed along the transport direction of the transport table and performing predetermined processing on a specimen accommodated in the analysis chip.
[0106] The disclosure of Japanese Patent Application No. 2022-192254 filed on Nov. 30, 2022 is incorporated in the present specification by reference. All documents, patent applications, and technical standards mentioned in the present specification are incorporated herein by reference to the same extent as in a case in which each document, each patent application, and each technical standard are specifically and individually described by being incorporated by reference.

Claims

1. A transport mechanism that transports an analysis chip including a body part having a flat bottom surface and a pair of engagement ribs provided to protrude outward from both ends in a width direction of the body part in a transport direction orthogonal to the width direction, the transport mechanism comprising: a transport table having a transport surface on which the analysis chip is placed with the bottom surface in contact with the transport surface, the analysis chip being transported while sliding on the transport surface; and a pair of screw shafts disposed parallel to each other along the transport direction, each of which has a shaft body and a helical protrusion helically formed on an outer peripheral surface of the shaft body, and the pair of screw shafts propels the analysis chip in the transport direction by rotating about their respective axes while the helical protrusions are engaged with the engagement ribs, wherein the transport surface has a hole formed in a contact region where the bottom surface of the analysis chip contacts the transport surface, and a recessed portion formed in at least a part of a periphery of an opening edge of the hole, and the recessed portion has a recessed surface that makes a height of a partial opening edge, which is located on a downstream side in the transport direction on an entire circumference of the opening edge and includes a contact point with a tangent extending in a direction orthogonal to the transport direction, lower than a height of the transport surface, and the recessed surface extends from the partial opening edge to an outside of the contact region.

2. The transport mechanism according to claim 1, wherein a ratio of a diameter of the hole to a length in the width direction of the body part is 8% or more, and the diameter is less than a length in the width direction of the body part.

3. The transport mechanism according to claim 1, wherein a diameter of the hole is 4 mm or more.

4. The transport mechanism according to claim 1, wherein the analysis chip has a planar shape in which a front end portion located on the downstream side in the transport direction has a front edge formed by a straight line extending in the direction orthogonal to the transport direction.

5. The transport mechanism according to claim 1, wherein a partial outer edge located on the downstream side in the transport direction on an entire circumference of an outer edge of the recessed portion and extending to the outside of the contact region extends toward the outside of the contact region in an inclined manner toward the downstream side in the transport direction with respect to the tangent, and the partial outer edge is a straight line or has an arc shape convex toward the outside of the contact region.

6. The transport mechanism according to claim 5, wherein an overall planar shape of the recessed portion including the hole is an oval shape of which a minor diameter is a diameter of the hole and one side in a longitudinal direction is the partial outer edge, and an entirety of a semicircular portion on an opposite side of the hole in the oval shape is located outside the contact region.

7. The transport mechanism according to claim 1, wherein a plurality of the holes are provided, and the recessed portion is provided for each of the holes.

8. The transport mechanism according to claim 1, wherein the analysis chip is a microchannel device including a microchannel for analyzing a specimen consisting of a biological sample.

9. An analysis apparatus comprising: the transport mechanism according to claim 1; and a plurality of processing units disposed along the transport direction of the transport table and performing predetermined processing on a specimen accommodated in the analysis chip.